

APHHS1005QWF/D
1.0 x 0.5 mm SMD Chip LED Lamp

DESCRIPTIONS

- The source color devices are made with InGaN Light Emitting Diode
- Electrostatic discharge and power surge could damage the LEDs
- It is recommended to use a wrist band or anti-electrostatic glove when handling the LEDs
- All devices, equipments and machineries must be electrically grounded

FEATURES

- 1.0 mm x 0.5 mm SMD LED, 0.5 mm thickness
- Low power consumption
- Ideal for backlight and indicator
- Package: 2000 pcs / reel
- Moisture sensitivity level: 3
- Halogen-free
- RoHS compliant

APPLICATIONS

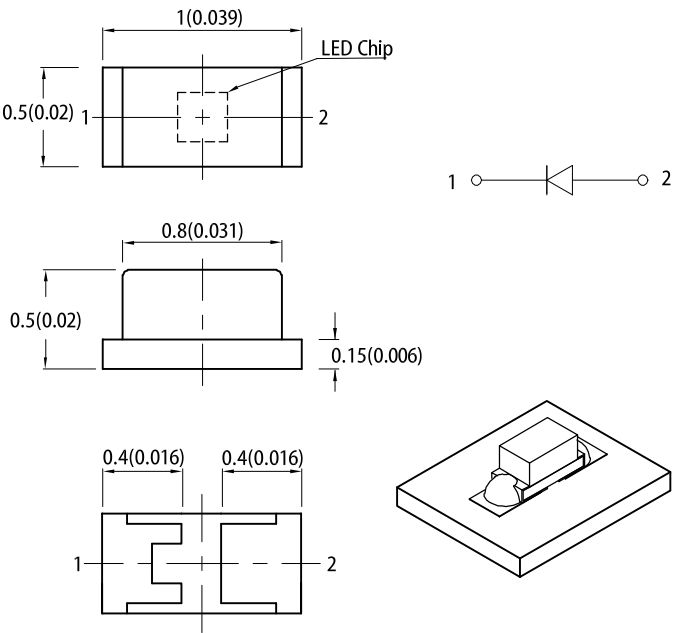
- Backlight
- Status indicator
- Home and smart appliances
- Wearable and portable devices
- Healthcare applications

ATTENTION

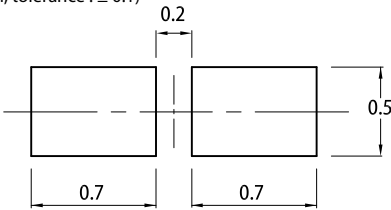
Observe precautions for handling electrostatic discharge sensitive devices



PACKAGE DIMENSIONS



RECOMMENDED SOLDERING PATTERN
(units : mm; tolerance : ± 0.1)



- Notes:
1. All dimensions are in millimeters (inches).
 2. Tolerance is ±0.1(0.004") unless otherwise noted.
 3. The specifications, characteristics and technical data described in the datasheet are subject to change without prior notice.
 4. The device has a single mounting surface. The device must be mounted according to the specifications.

SELECTION GUIDE

Part Number	Emitting Color (Material)	Lens Type	Iv (mcd) @ 20mA ^[2]		Viewing Angle ^[1]
			Min.	Typ.	2θ1/2
APHHS1005QWF/D	White (InGaN)	Yellow Fluorescent	120	200	160°

Notes:

1. θ1/2 is the angle from optical centerline where the luminous intensity is 1/2 of the optical peak value.
2. Luminous intensity / luminous flux: +/-15%.
3. Luminous intensity value is traceable to CIE127-2007 standards.

ELECTRICAL / OPTICAL CHARACTERISTICS at $T_A=25^{\circ}\text{C}$

Parameter	Symbol	Emitting Color	Value		Unit
			Typ.	Max.	
Chromaticity Coordinates x $I_F = 20\text{mA}$	$x^{[1]}$	White	0.31	-	-
Chromaticity Coordinates y $I_F = 20\text{mA}$	$y^{[1]}$	White	0.31	-	-
Forward Voltage $I_F = 20\text{mA}$	$V_F^{[2]}$	White	3.3	4.0	V
Reverse Current ($V_R = 5\text{V}$)	I_R	White	-	50	μA
Temperature Coefficient of x $I_F = 20\text{mA}$, $-10^{\circ}\text{C} \leq T \leq 85^{\circ}\text{C}$	TC_x	White	-0.18	-	$10^{-3}/^{\circ}\text{C}$
Temperature Coefficient of y $I_F = 20\text{mA}$, $-10^{\circ}\text{C} \leq T \leq 85^{\circ}\text{C}$	TC_y	White	-0.19	-	$10^{-3}/^{\circ}\text{C}$
Temperature Coefficient of V_F $I_F = 20\text{mA}$, $-10^{\circ}\text{C} \leq T \leq 85^{\circ}\text{C}$	TC_V	White	-3.0	-	$\text{mV}/^{\circ}\text{C}$

Notes:

1. Measurement tolerance of the chromaticity coordinates is ± 0.01 .2. Forward voltage: $\pm 0.1\text{V}$.

3. Excess driving current and / or operating temperature higher than recommended conditions may result in severe light degradation or premature failure.

ABSOLUTE MAXIMUM RATINGS at $T_A=25^{\circ}\text{C}$

Parameter	Symbol	Value	Unit
Power Dissipation	P_D	120	mW
Reverse Voltage	V_R	5	V
Junction Temperature	T_j	115	$^{\circ}\text{C}$
Operating Temperature	T_{op}	-40 to +85	$^{\circ}\text{C}$
Storage Temperature	T_{stg}	-40 to +85	$^{\circ}\text{C}$
DC Forward Current	I_F	30	mA
Peak Forward Current	$I_{FP}^{[1]}$	150	mA
Electrostatic Discharge Threshold (HBM)	-	250	V
Thermal Resistance (Junction / Ambient)	$R_{th JA}^{[2]}$	700	$^{\circ}\text{C}/\text{W}$
Thermal Resistance (Junction / Solder point)	$R_{th JS}^{[2]}$	600	$^{\circ}\text{C}/\text{W}$

Notes:

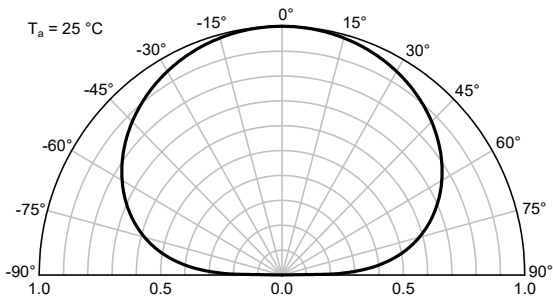
1. 1/10 Duty Cycle, 0.1ms Pulse Width.

2. $R_{th JA}$, $R_{th JS}$ Results from mounting on PC board FR4 (pad size $\geq 16\text{ mm}^2$ per pad).

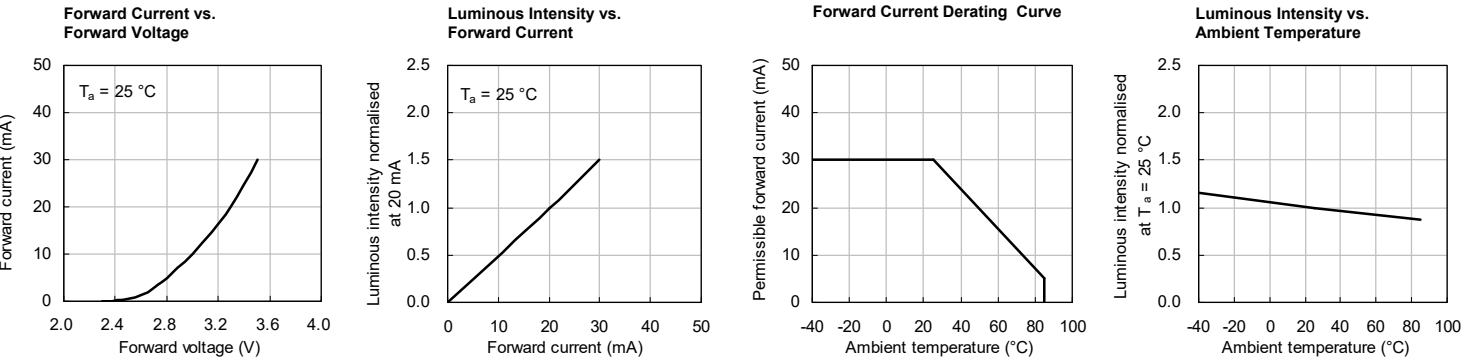
3. Relative humidity levels maintained between 40% and 60% in production area are recommended to avoid the build-up of static electricity – Ref JEDEC/JESD625-A and JEDEC/J-STD-033.

TECHNICAL DATA

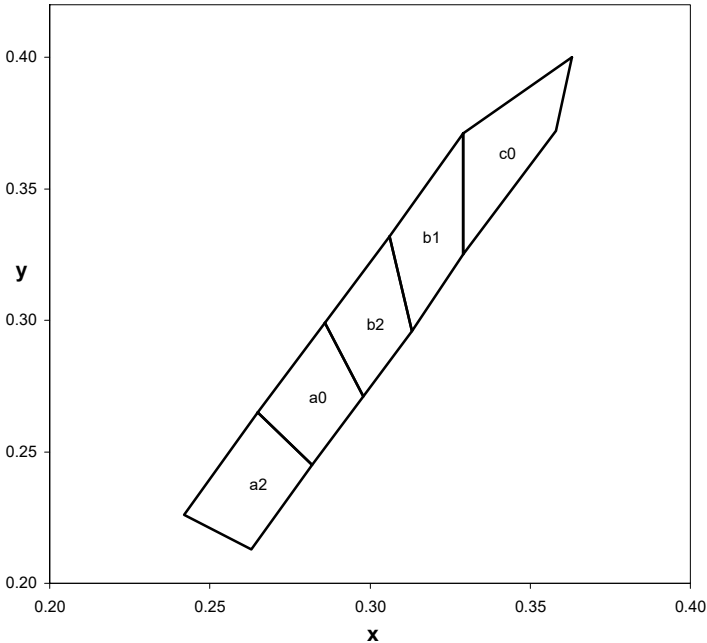
SPATIAL DISTRIBUTION



WHITE



CIE CHROMATICITY DIAGRAM

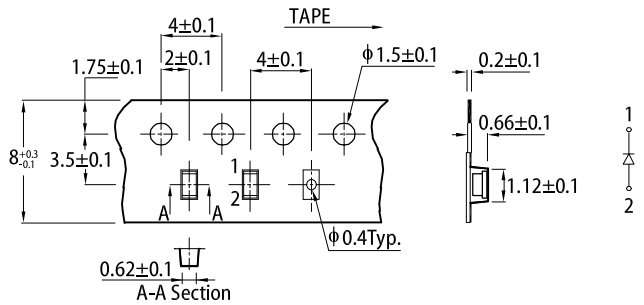
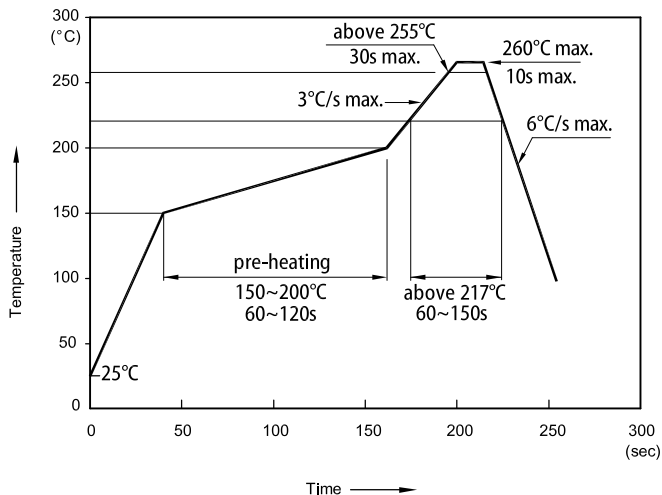


x		y	x		y
a2	0.263	0.213	a0	0.282	0.245
	0.282	0.245		0.298	0.271
	0.265	0.265		0.286	0.299
	0.242	0.226		0.265	0.265
b2	0.298	0.271	b1	0.313	0.296
	0.313	0.296		0.329	0.325
	0.306	0.332		0.329	0.371
	0.286	0.299		0.306	0.332
c0	0.329	0.325			
	0.358	0.372			
	0.363	0.400			
	0.329	0.371			

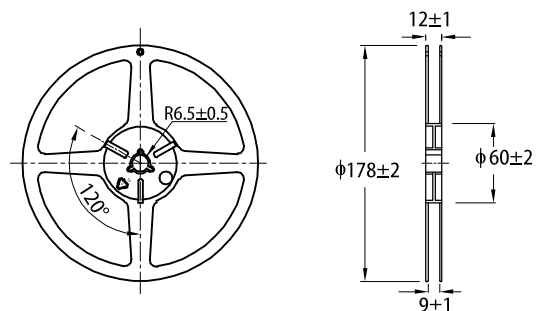
Notes:
Shipment may contain more than one chromaticity regions.
Orders for single chromaticity region are generally not accepted.
Measurement tolerance of the chromaticity coordinates is ± 0.01 .

REFLOW SOLDERING PROFILE for LEAD-FREE SMD PROCESS

TAPE SPECIFICATIONS (units : mm)

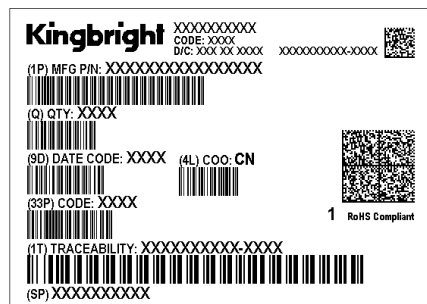
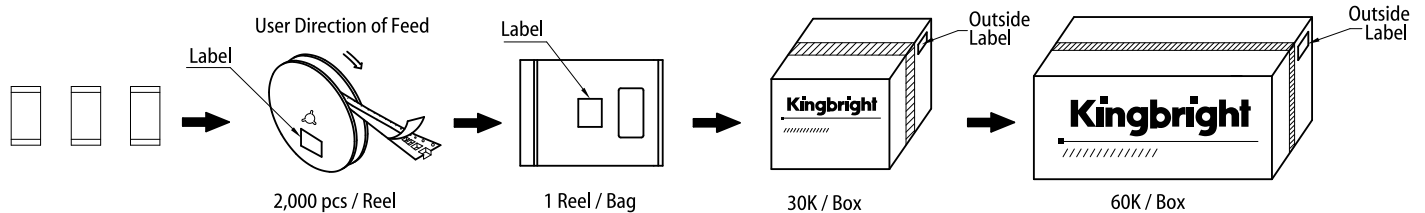


REEL DIMENSION (units : mm)



- Notes:
1. Don't cause stress to the LEDs while it is exposed to high temperature.
 2. The maximum number of reflow soldering passes is 2 times.
 3. Reflow soldering is recommended. Other soldering methods are not recommended as they might cause damage to the product.

PACKING & LABEL SPECIFICATIONS



PRECAUTIONARY NOTES

1. The information included in this document reflects representative usage scenarios and is intended for technical reference only.
2. The part number, type, and specifications mentioned in this document are subject to future change and improvement without notice. Before production usage customer should refer to the latest datasheet for the updated specifications.
3. When using the products referenced in this document, please make sure the product is being operated within the environmental and electrical limits specified in the datasheet. If customer usage exceeds the specified limits, Kingbright will not be responsible for any subsequent issues.
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